



US012408322B2

(12) **United States Patent**
Jang et al.

(10) **Patent No.:** **US 12,408,322 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SEMICONDUCTOR STRUCTURE AND METHOD FOR MANUFACTURING THE SAME**

(71) Applicant: **CHANGXIN MEMORY TECHNOLOGIES, INC.**, Hefei (CN)

(72) Inventors: **Semyeong Jang**, Hefei (CN); **Joonsuk Moon**, Hefei (CN); **Deyuan Xiao**, Hefei (CN); **Minki Hong**, Hefei (CN); **Kyongtaek Lee**, Hefei (CN); **Jo-Lan Chin**, Hefei (CN)

(73) Assignee: **CHANGXIN MEMORY TECHNOLOGIES, INC.**, Hefei (CN)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 392 days.

(21) Appl. No.: **17/935,567**

(22) Filed: **Sep. 26, 2022**

(65) **Prior Publication Data**

US 2023/0014884 A1 Jan. 19, 2023

Related U.S. Application Data

(63) Continuation of application No. PCT/CN2022/102663, filed on Jun. 30, 2022.

(51) **Int. Cl.**
H10B 12/00 (2023.01)
G11C 5/06 (2006.01)

(52) **U.S. Cl.**
CPC **H10B 12/315** (2023.02); **G11C 5/063** (2013.01); **H10B 12/05** (2023.02)

(58) **Field of Classification Search**
CPC .. **H10B 12/315**; **H10B 12/05**; **H10B 12/0335**; **G11C 5/063**

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,244,824 A	9/1993	Sivan
6,376,303 B1	4/2002	Seo
8,293,604 B2	10/2012	Yoon
9,673,257 B1	6/2017	Takaki
2006/0043471 A1	3/2006	Tang
2006/0046391 A1	3/2006	Tang

(Continued)

FOREIGN PATENT DOCUMENTS

CN	101017825 A	8/2007
CN	101740500 A	6/2010

(Continued)

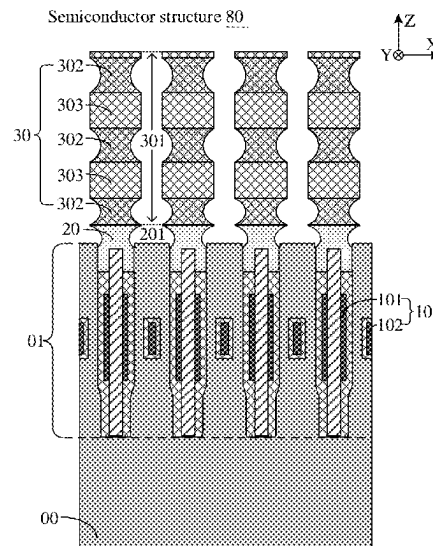
Primary Examiner — Jonathan Han

(74) *Attorney, Agent, or Firm* — SYNCODA LLC; Feng Ma

(57) **ABSTRACT**

A semiconductor structure includes a base, a dielectric layer, a gate structure, and a covering layer. The base includes discrete semiconductor pillars. The semiconductor pillars are disposed at the top of the base and extend in a vertical direction. The dielectric layer covers the sidewall of the semiconductor pillar. The gate structure is disposed in the middle area of the semiconductor pillar. The gate structure includes a gate-all-around structure, the gate-all-around surrounding the semiconductor pillar. A first part of the dielectric layer is disposed between the gate structures and the semiconductor pillars. The covering layer covers the top of the semiconductor pillar and part of the sidewall close to the top. The material of the covering layer includes a boron-containing compound.

15 Claims, 14 Drawing Sheets



(56)

References Cited**U.S. PATENT DOCUMENTS**

2007/0020819	A1	1/2007	Tang	
2007/0048943	A1	3/2007	Tang	
2007/0181925	A1	8/2007	Yoon	
2007/0224753	A1	9/2007	Tang	
2009/0072291	A1	3/2009	Takaishi	
2009/0207649	A1	8/2009	Tang	
2010/0120221	A1 *	5/2010	Kang	H10D 30/63 438/430
2010/0213525	A1	8/2010	Masuoka	
2011/0104862	A1	5/2011	Kadoya	
2011/0171796	A1	7/2011	Tang	
2013/0011987	A1 *	1/2013	Park	H10D 30/025 438/585
2013/0056698	A1	3/2013	Satoh	
2013/0075813	A1	3/2013	Kadoya	
2015/0017767	A1	1/2015	Masuoka et al.	
2015/0380548	A1	12/2015	Wang et al.	
2016/0204251	A1	7/2016	Masuoka et al.	
2017/0271510	A1	9/2017	Wang et al.	
2017/0309632	A1	10/2017	Masuoka et al.	
2017/0323977	A1	11/2017	Cheng et al.	
2018/0083136	A1	3/2018	Xie et al.	
2018/0096896	A1	4/2018	Zhu	
2018/0175212	A1	6/2018	Cheng et al.	
2019/0027570	A1	1/2019	Ching et al.	
2019/0115438	A1	4/2019	Ching et al.	
2019/0123053	A1	4/2019	Masuoka et al.	
2019/0237581	A1	8/2019	Saito et al.	

2019/0326395	A1	10/2019	Ando et al.	
2020/0020812	A1	1/2020	Masuoka et al.	
2020/0052084	A1	2/2020	Ching et al.	
2021/0305431	A1	9/2021	Ishimaru et al.	
2022/0069070	A1	3/2022	Lai et al.	
2022/0102347	A1	3/2022	Lai et al.	
2022/0139918	A1	5/2022	Lee	
2022/0139920	A1	5/2022	Lee	
2022/0199837	A1	6/2022	Masuoka et al.	
2022/0262954	A1	8/2022	Ishimaru et al.	
2023/0100189	A1 *	3/2023	Kim	H10D 30/6757 257/401
2023/0335436	A1 *	10/2023	Wang	H01L 23/5226

FOREIGN PATENT DOCUMENTS

CN	101933135	A	12/2010
CN	102867752	A	1/2013
CN	105280698	A	1/2016
CN	106252352	A	12/2016
CN	107845578	A	3/2018
CN	110476230	A	11/2019
CN	110931429	A	3/2020
CN	113078155	A	7/2021
CN	113078156	A	7/2021
CN	113451405	A	9/2021
CN	114464535	A	5/2022
CN	114497039	A	5/2022
EP	1804286	A1	7/2007
WO	2013033267	A1	3/2013

* cited by examiner

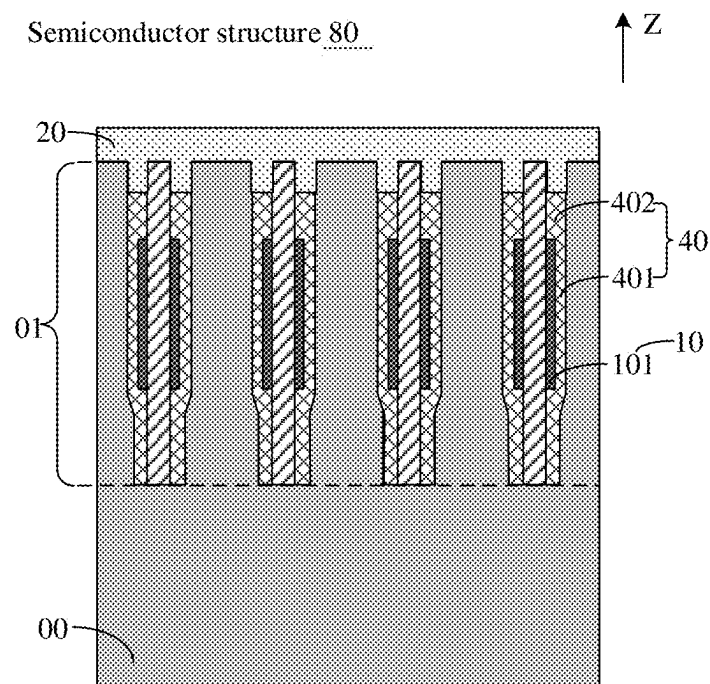


FIG. 1

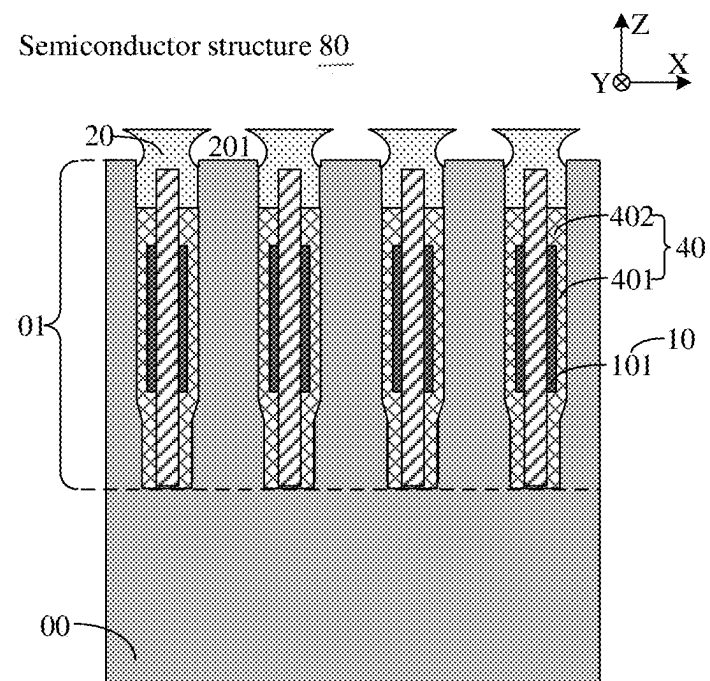


FIG. 2

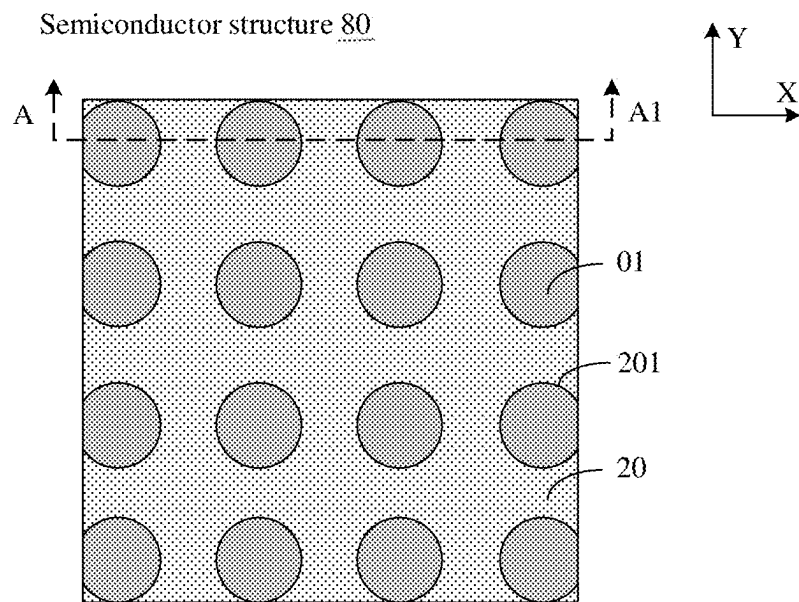


FIG. 3

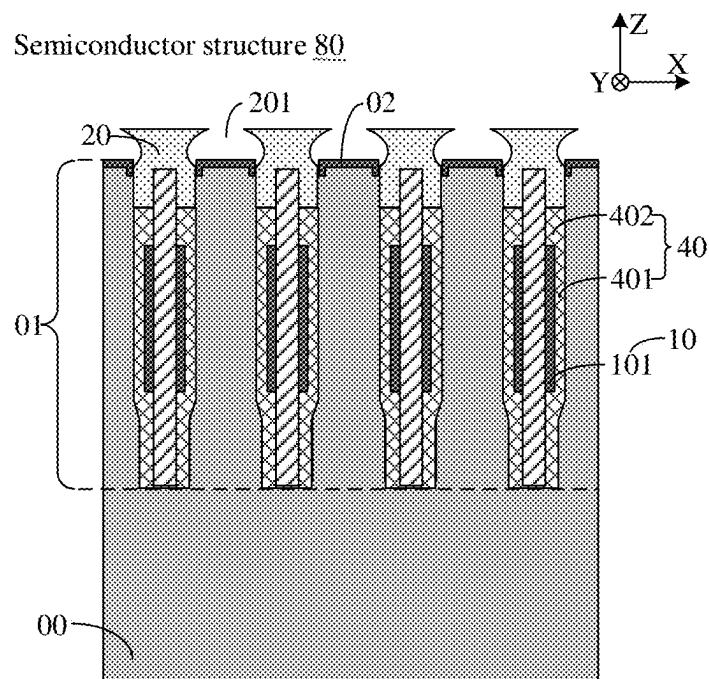


FIG. 4

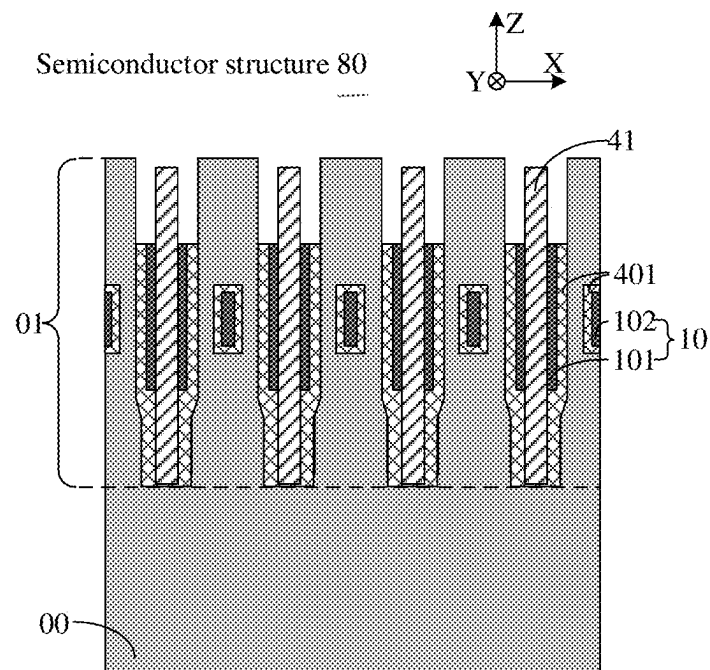


FIG. 5

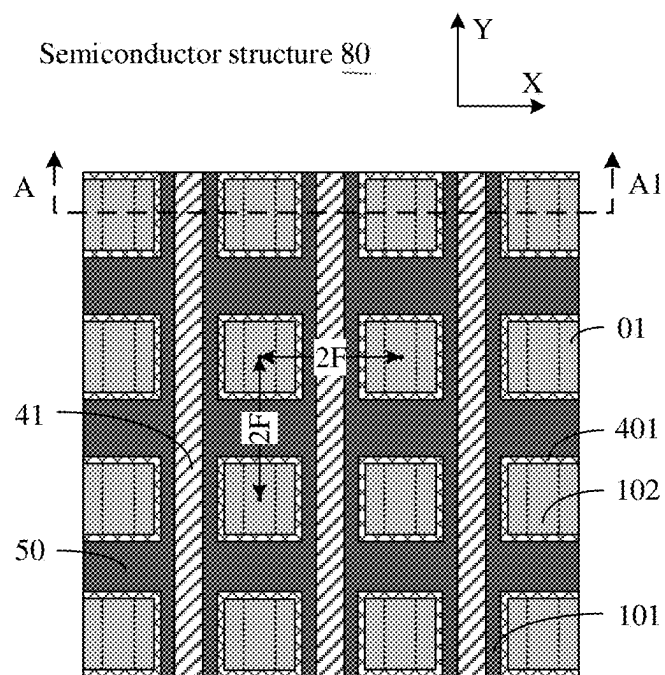


FIG. 6

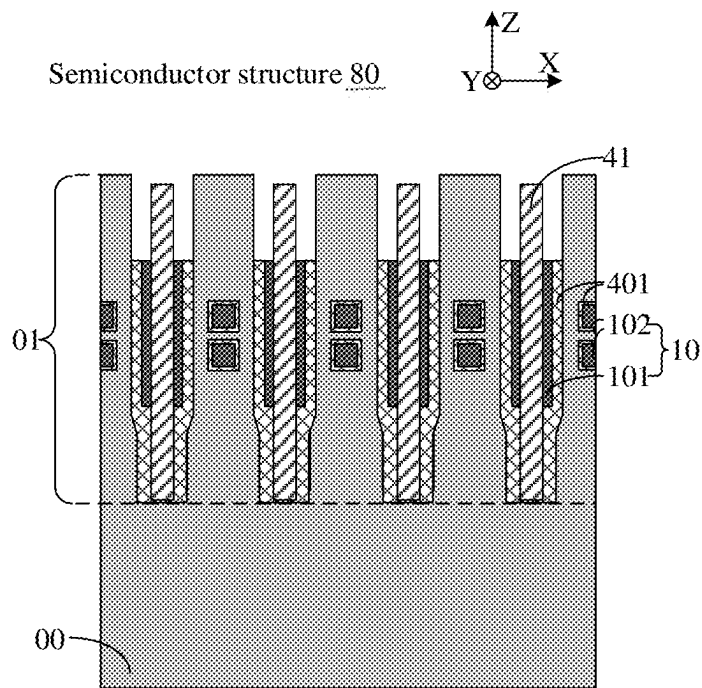


FIG. 7

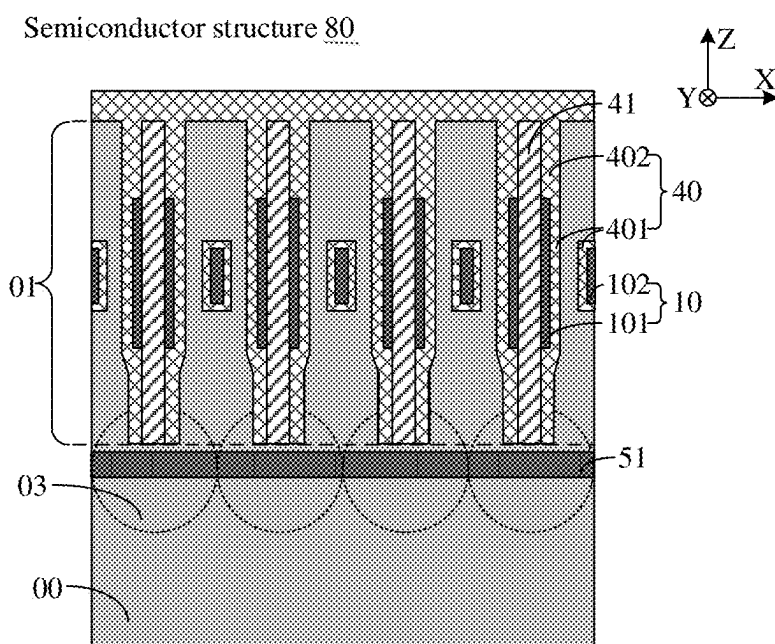
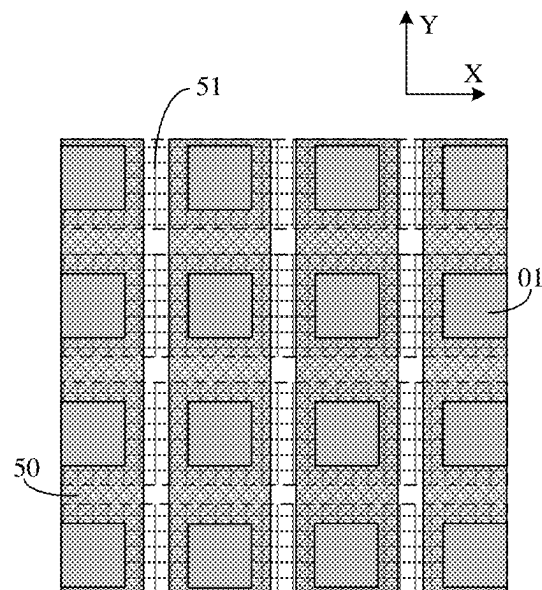


FIG. 8

**FIG. 9**

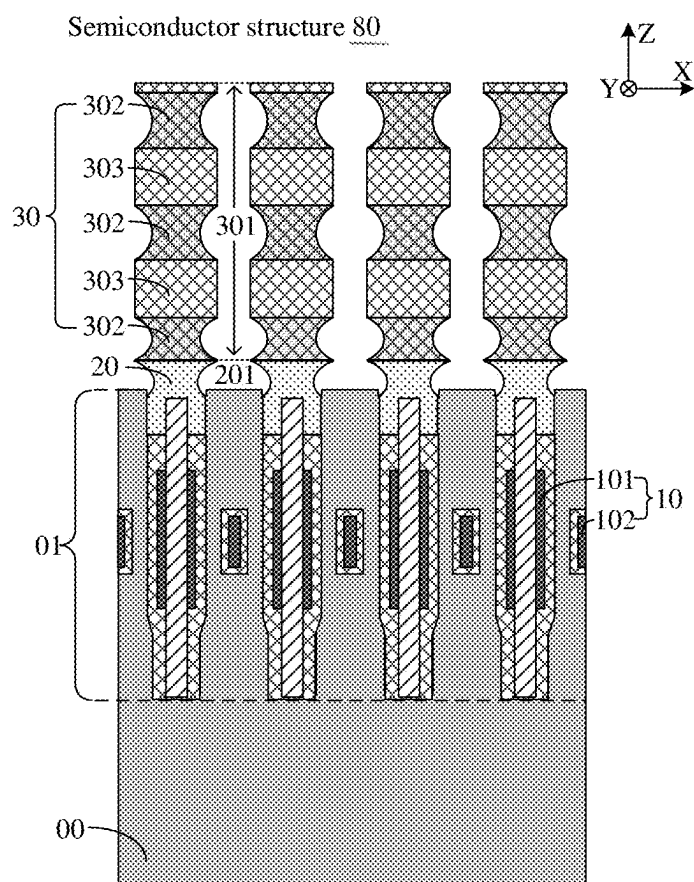


FIG. 10

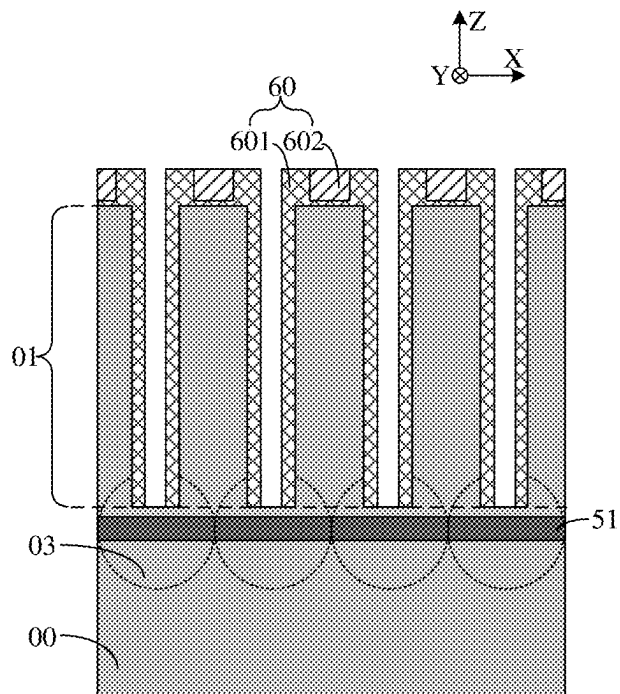


FIG. 11

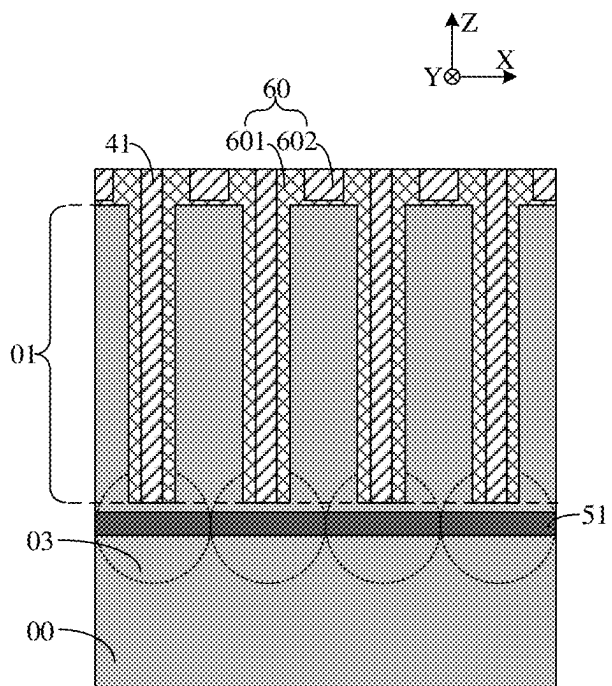


FIG. 12

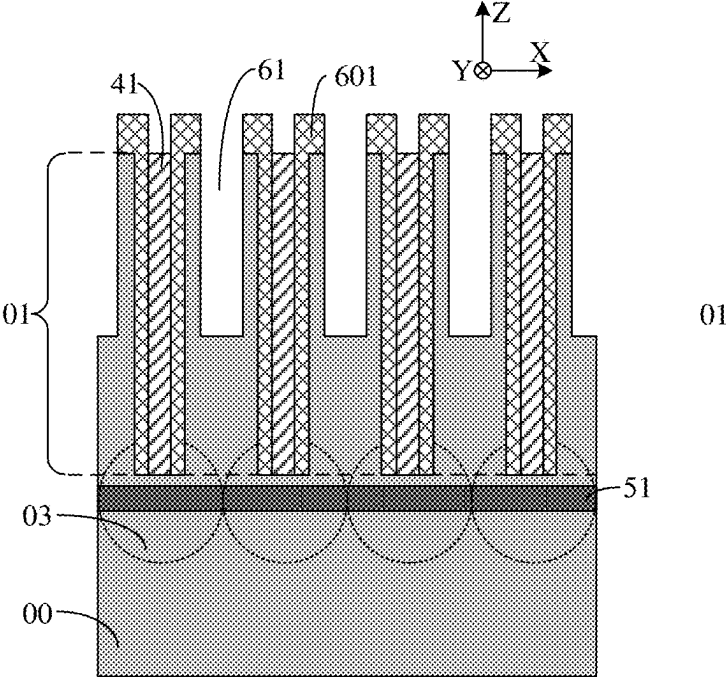


FIG. 13

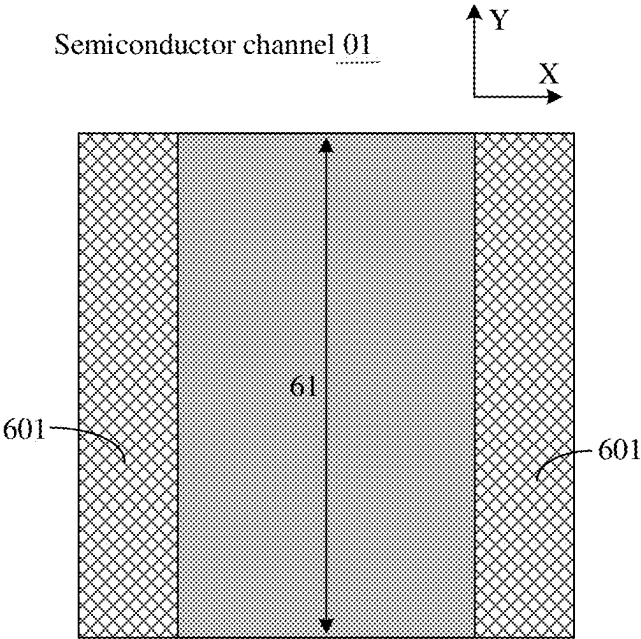


FIG. 14

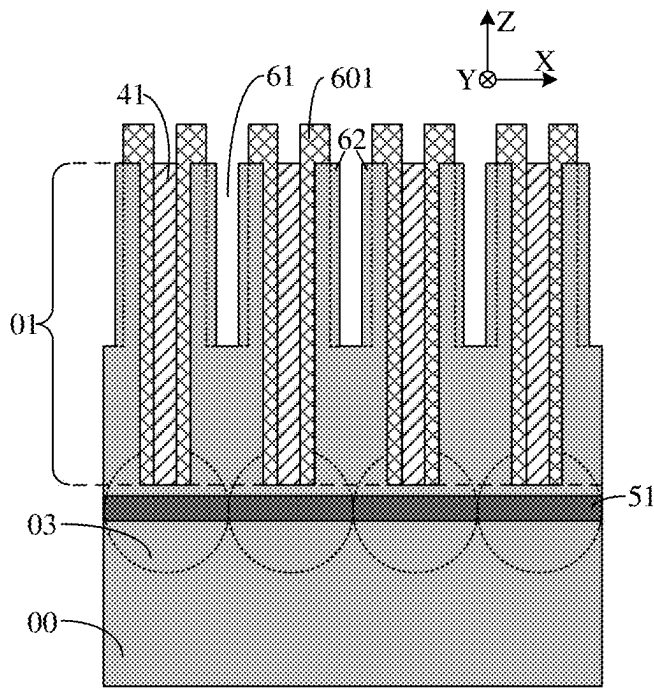


FIG. 15

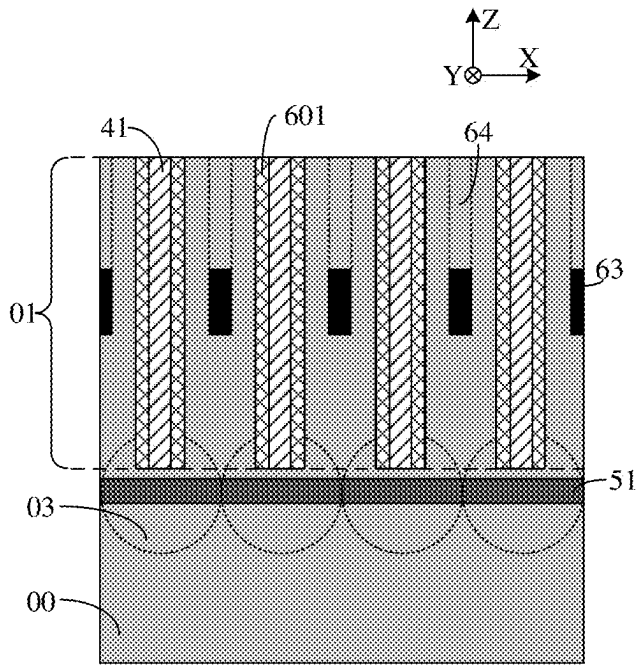


FIG. 16

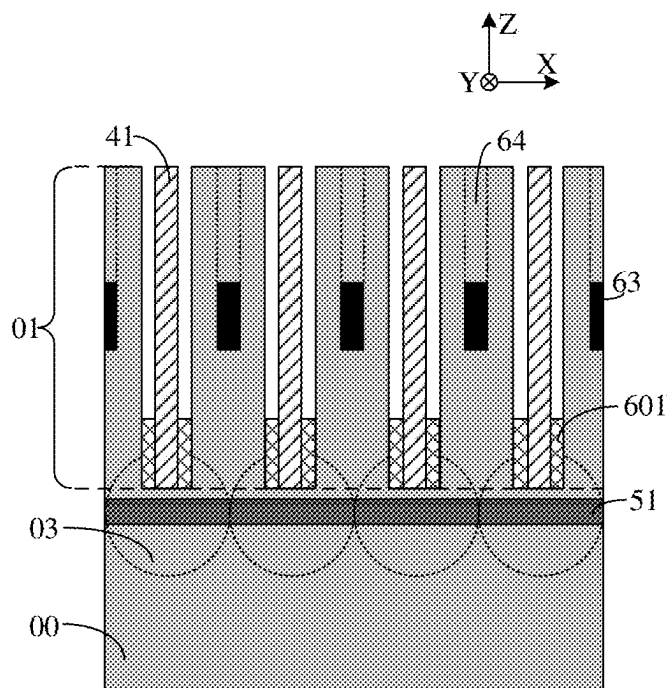


FIG. 17

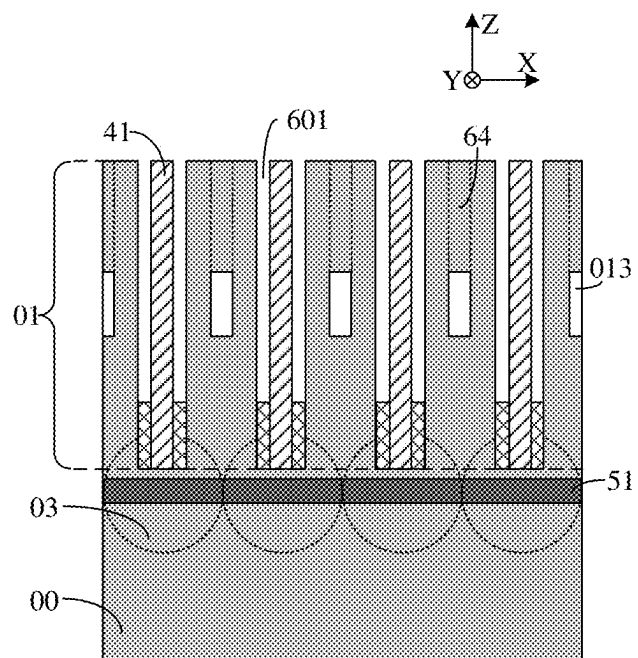


FIG. 18

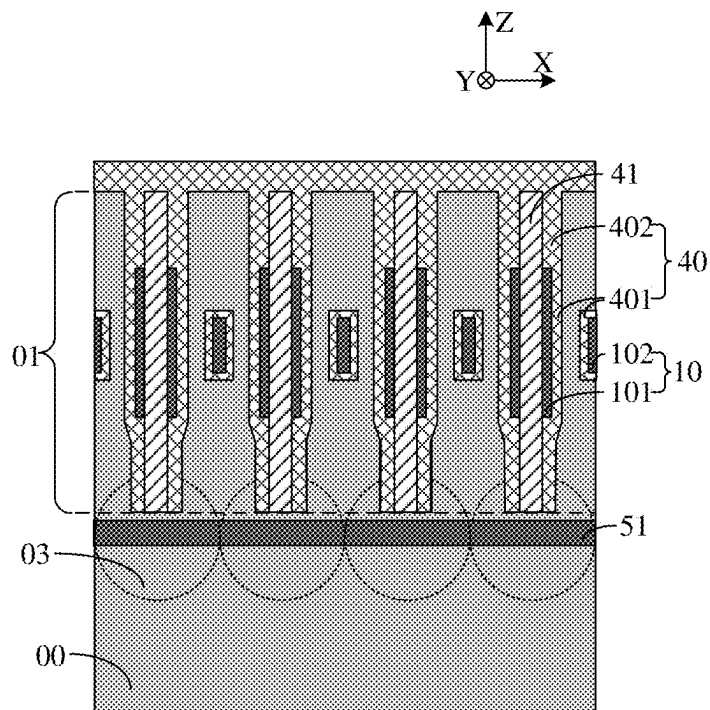


FIG. 19

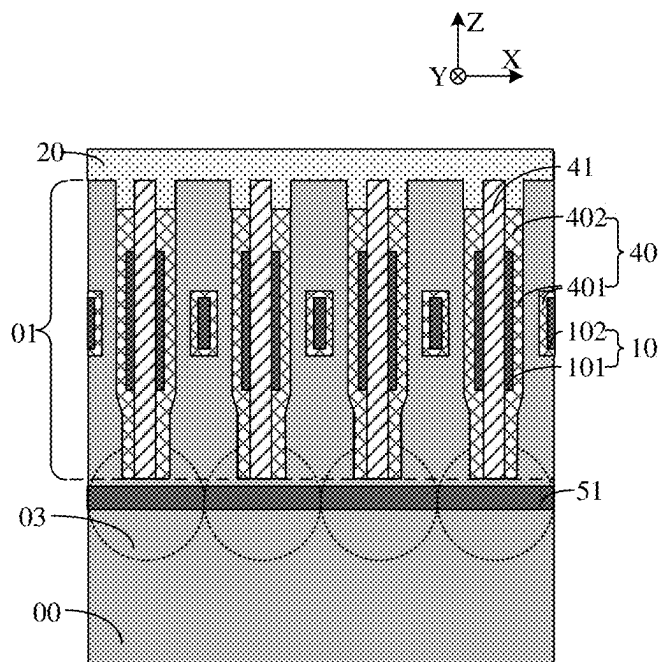


FIG. 20

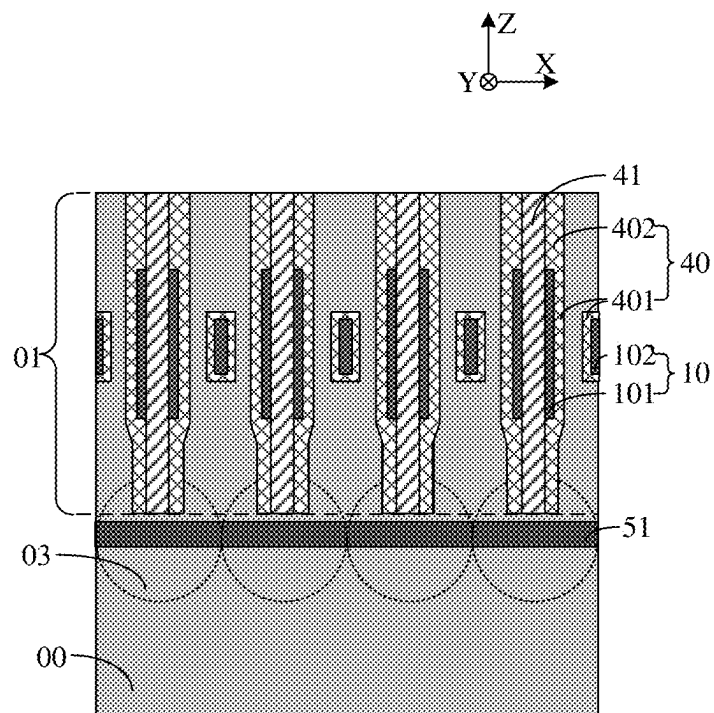


FIG. 21

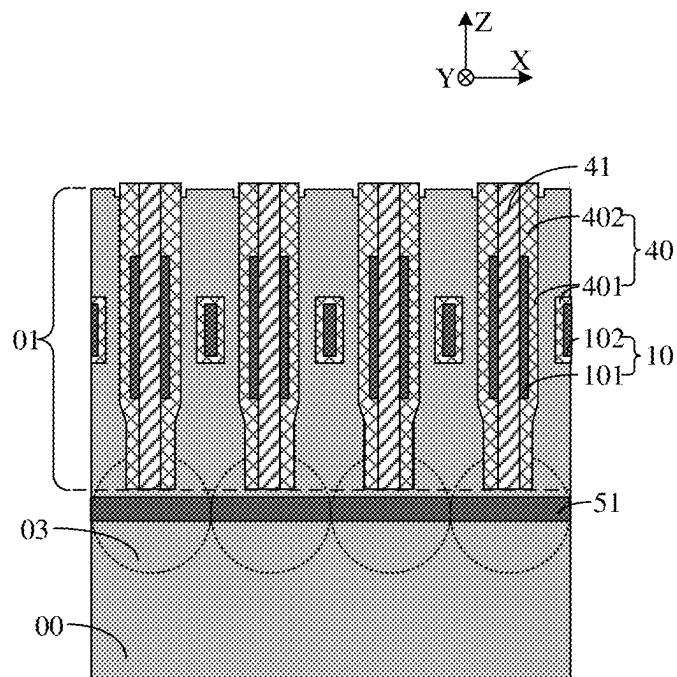


FIG. 22

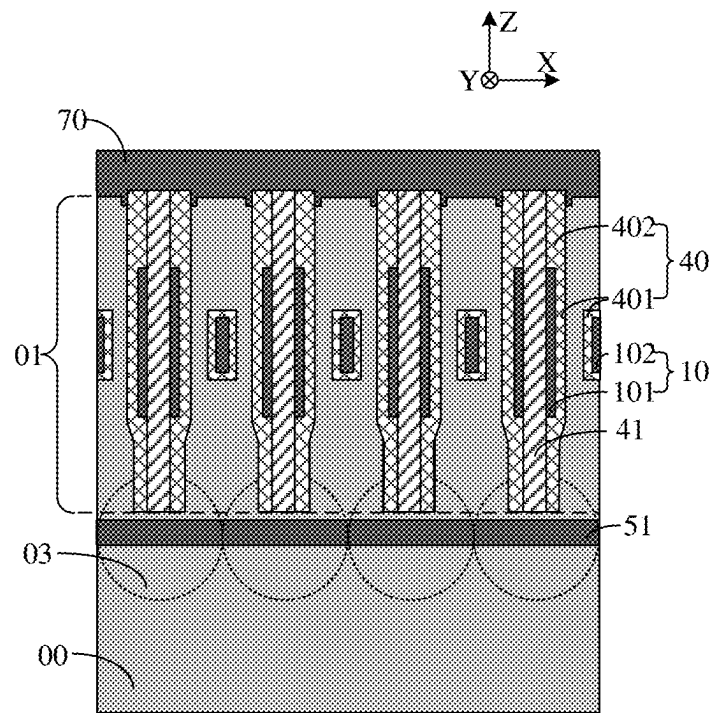


FIG. 23

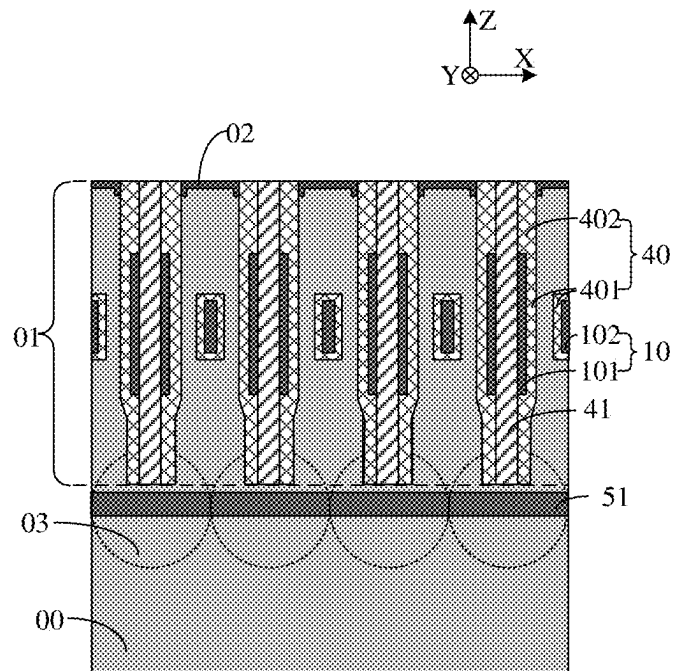


FIG. 24

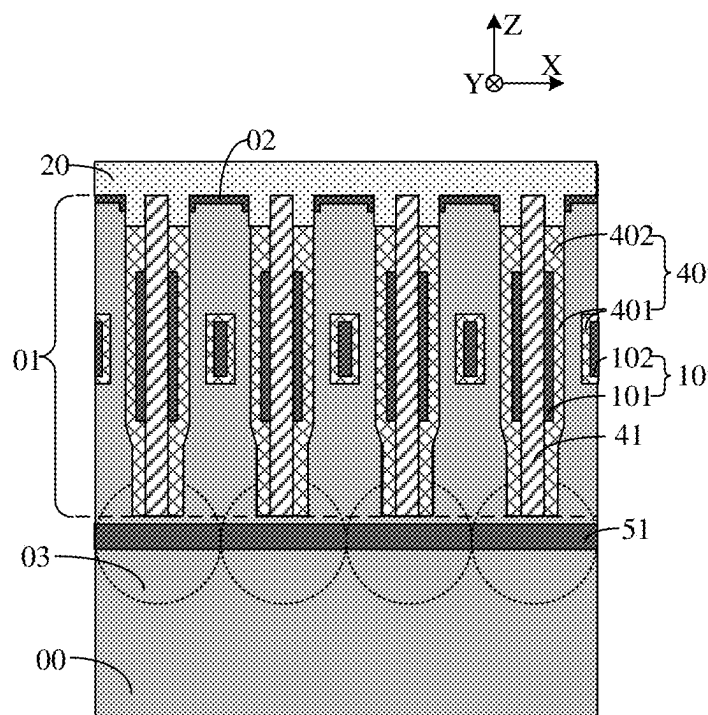


FIG. 25

1

SEMICONDUCTOR STRUCTURE AND METHOD FOR MANUFACTURING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

This is a continuation application of International Patent Application No. PCT/CN2022/102663 filed on Jun. 30, 2022, which claims priority to Chinese Patent Application No. 202210645321.1 filed on Jun. 8, 2022. The disclosures of the above-referenced applications are hereby incorporated by reference in their entirety.

BACKGROUND

As the integration density of a dynamic memory develops to be higher, while the arrangement mode of transistors in a dynamic memory array structure and how to reduce the size of a single functional device in the dynamic memory array structure are researched, the electrical performance of the small-size functional device needs to be improved.

Higher density efficiency may be achieved when a Vertical Gate All Around (VGAA) transistor structure is used as an access transistor of the dynamic memory. However, in some implementations, the electrical performance of a semiconductor structure related to the access transistor is low.

SUMMARY

In view of this, embodiments of the present disclosure provide a semiconductor structure and a method for manufacturing the same.

The technical solutions of the embodiments of the present disclosure are implemented as follows.

The embodiments of the present disclosure provide a semiconductor structure, which may include: a base, a dielectric layer, gate structures, and a covering layer.

The base may include discrete semiconductor pillars. The semiconductor pillars are disposed at top of the base and extend in a vertical direction.

The dielectric layer covers the sidewalls of the semiconductor pillars.

The gate structures are disposed in the middle area of the semiconductor pillars. The gate structure may include a gate-all-around structure, and the gate-all-around surrounds the semiconductor pillar. A first part of the dielectric layer is disposed between the gate structures and the semiconductor pillars.

The covering layer covers the top of the semiconductor pillars and part of the sidewalls close to the top. The material of the covering layer may include a boron-containing compound.

In the above solution, the material of the covering layer may include a silicon boron material.

In the above solution, the silicon boron material is silicon boron nitride SiB_xN_y , and $y-x \leq 2$.

In the above solution, the boron silicon material is Boro Phospho Silicate Glass (BPSG) or boro silicate glass.

In the above solution, the gate structures may further include at least one bridge gate structure. The at least one bridge gate structure penetrates through the semiconductor pillars and extends to the inner walls of the gate-all-around structures in the penetrating direction. The at least one bridge gate structures are disposed in the semiconductor pillars and are disposed corresponding to the middle area of the gate-all-around structures.

2

In the above solution, the width of each bridge gate structure is less than that of the semiconductor pillar.

In the above solution, the gate structure may further include at least two bridge gate structures. The at least two bridge gate structures are stacked at an interval in the vertical direction, and the at least two bridge gate structures are isolated directly by the semiconductor pillar.

In the above solution, the top of the gate-all-around structure is lower than the top of the semiconductor pillar. The second part of the dielectric layer covers the top of the gate-all-around structure and is disposed between the gate-all-around structure and the covering layer. The thickness of the second part of the dielectric layer is greater than that of the first part of the dielectric layer.

In the above solution, the semiconductor structure may further include: a first isolation layer. The first isolation layer is disposed between the adjacent semiconductor pillars, and the gate structure is disposed between the first isolation layer and the semiconductor pillar.

In the above solution, the semiconductor structure may further include: a bit line. The bit line is disposed in the base. The bottom of the semiconductor pillar is connected with the bit line.

The embodiments of the present disclosure also provide a method for manufacturing a semiconductor structure, which may include the following operations.

A base is provided, the base is patterned, and discrete semiconductor pillars are formed on the base. The semiconductor pillars are disposed at the top of the base and extend in a vertical direction.

A dielectric layer is formed. The dielectric layer covers the sidewalls of the semiconductor pillars.

A gate structures are formed in the middle area of the semiconductor pillars. The gate structure may include a gate-all-around structure, and the gate-all-around surrounds the semiconductor pillar. A first part of the dielectric layer is disposed between the gate structures and the semiconductor pillars.

A covering layer is formed. The covering layer covers the top of the semiconductor pillars and part of the sidewalls close to the top. The material of the covering layer may include a boron-containing compound.

In the above solution, the gate structure may further include one bridge gate structure, the step of forming the gate structure may include that: a groove is formed in the semiconductor pillar; a sacrificial layer is formed in the groove; and a filling layer is formed in the groove, the filling layer is disposed on the sacrificial layer, and the remaining groove is filled with the filling layer.

In the above solution, the gate structure may further include at least two bridge gate structures, the step of forming the gate structure may include that: a groove is formed in the semiconductor pillar; a sacrificial layer is formed in the groove; part of the sacrificial layer is etched back, and a filling layer is formed in the groove, the filling layer being disposed on the remaining sacrificial layer; and the sacrificial layer and the filling layer are continued to be formed in the groove in sequence, and the top of the groove is filled with the filling layer.

In the above solution, the step of forming the gate structure may further include that: the sacrificial layer is removed to form a through hole: the through hole penetrates through the semiconductor pillar; a first initial dielectric layer is formed on the sidewall of the semiconductor pillar and the inner wall of the through hole: the part, disposed in the middle area of the semiconductor pillar, of the first initial dielectric layer forms the first part of the dielectric layer; the

3

gate layer is formed around the sidewall of the initial dielectric layer through filling the through hole, and the part of the gate layer for filling the through hole forms a bridge gate structure; and the part of the gate layer around the sidewall of the first initial dielectric layer is etched to form the gate-all-around structure.

In the above solution, after the through hole is formed and before the first initial dielectric layer is formed, the manufacturing method may further include that: part of the sidewall of the semiconductor pillar is etched, so that the width of the middle area of the semiconductor pillar is smaller than the width of the bottom of the semiconductor pillar.

In the above solution, the step of forming the covering layer may include that: a second initial dielectric layer is formed on the gate structure and the semiconductor pillar: the second initial dielectric layer is etched until the top of the semiconductor pillar and part of the sidewall close to the top are exposed: the remaining second initial dielectric layer forms a second part of the dielectric layer; and the top of the semiconductor pillar and part of the sidewall close to the top are covered to form the covering layer.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic diagram I of a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 2 is a schematic diagram II of a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 3 is a schematic diagram III of a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 4 is a schematic diagram IV of a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 5 is a schematic diagram V of a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 6 is a schematic diagram VI of a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 7 is a schematic diagram VII of a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 8 is a schematic diagram VIII of a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 9 is a schematic diagram IX of a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 10 is a schematic diagram X of a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 11 is a schematic diagram I of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 12 is a schematic diagram II of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 13 is a schematic diagram III of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 14 is a schematic diagram IV of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

4

FIG. 15 is a schematic diagram V of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 16 is a schematic diagram VI of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 17 is a schematic diagram VII of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 18 is a schematic diagram VIII of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 19 is a schematic diagram IX of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 20 is a schematic diagram X of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 21 is a schematic diagram XI of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 22 is a schematic diagram XII of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 23 is a schematic diagram XIII of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 24 is a schematic diagram XIV of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

FIG. 25 is a schematic diagram XV of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

DETAILED DESCRIPTION

In order to make the purposes, technical solutions and advantages of the present disclosure clearer, the technical solutions of the present disclosure are further described in details below in combination with the accompanying drawings and the embodiments. The described embodiments should not be regarded as limitations of the present disclosure. All other embodiments obtained by those of ordinary skill in the art without creative work shall fall within the scope of protection of the present disclosure.

In the following description, reference is made to “some embodiments” which describe a subset of all possible embodiments, but it can be understood that “some embodiments” may be the same or a different subset of all possible embodiments, and may be combined with each other without conflict.

If a similar description of “first/second” appears in the application document, the following description will be added. In the following description, the involved terms “first/second/third” are only used to distinguish similar objects, and do not represent a specific order of the objects. It can be understood that the specific order or sequence of “first/second/third” may be interchangeable under the allowable circumstances, so that the embodiments of the present disclosure described herein may be implemented in an order other than those illustrated or described herein.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the art of the present disclosure. The terms used herein is only for the purpose of describing the embodiments of the present disclosure and is not intended to limit the present disclosure.

5

FIG. 1 is an optional schematic structural diagram of a semiconductor structure provided by an embodiment of the present disclosure, as shown in FIG. 1, the semiconductor structure 80 includes: a base 00, gate structures 10, a covering layer 20, and a dielectric layer 40.

The base 00 includes discrete semiconductor pillars 01. The semiconductor pillars 01 are disposed at the top of the base 00 and extend in a vertical direction Z.

The dielectric layer 40 covers the sidewalls of the semiconductor pillars 01.

The gate structures 10 are disposed in the middle area of the semiconductor pillars 01. The gate structure 10 includes a gate-all-around structure 101, the gate-all-around 101 surrounds the semiconductor pillar 01. A first part 401 of the dielectric layer 40 is disposed between the gate structure 10 and the semiconductor pillar 01.

The covering layer 20 covers the top of the semiconductor pillar 01 and part of the sidewall close to the top. The material of the covering layer 20 includes a boron-containing compound.

In the embodiment of the present disclosure, the base 00 may include at least one of semiconductor materials, for example, group IV elements such as silicon (Si), germanium (Ge), and silicon germanium (SiGe), or group III-V components such as gallium arsenide (GaAs), aluminum gallium arsenide (AlGaAs), indium arsenide (InAs), or indium gallium arsenide (InGaAs), and the following is exemplarily illustrated with the silicon element included in the base 00.

The semiconductor pillar 01 may have a doping element to improve the conductivity of the semiconductor pillar 01. Herein, the doping element may be a P-type doping element or an N-type doping element, and the N-type doping element may be at least one of an arsenic (As) element, a phosphorus (P) element, or an antimony (Sb) element. The P-type doping element may be at least one of a boron (B) element, an indium (In) element, or a gallium (Ga) element.

The material of the gate structure 10 may be titanium nitride (TiN), or at least one of conductive materials such as polycrystalline silicon, tantalum nitride (TaN), copper (Cu), or tungsten (W), and the following is exemplarily illustrated with titanium nitride. The material of the dielectric layer 40 may be silicon oxide (SiO).

In the embodiment of the present disclosure, the single semiconductor pillar 01, the gate structure 10 surrounds the semiconductor pillar 01, and the first part 401 of the dielectric layer 40 disposed between the semiconductor pillars 01 and the gate structures 10 may together form a first transistor. Herein, a channel area is formed in the middle area of the semiconductor pillar 01 corresponding to the gate structure 10, the gate structure 10 is used as a gate of the first transistor, and the parts 401 of the upper and lower ends of the middle area of the semiconductor pillar 01 respectively form a source and a drain. It is to be noted that the middle area of the semiconductor pillar 01 is the middle part of the semiconductor pillar 01 in the vertical direction Z, and the middle area of the semiconductor pillar 01 is between the top and the bottom of the semiconductor pillar 01 and has a certain distance from the top and the bottom. The first part 401 of the dielectric layer 40 is used as a gate dielectric of the first transistor.

The first transistor is a VGAA transistor, that is, the channel formed by the semiconductor pillar 01 extends in the vertical direction, and the gate-all-around structure 101 of the gate structure 10 surrounds the channel. Therefore, under the same size, compared with the transistor structures such as a Fin Field-Effect Transistor (FinFET), etc., the gate

6

of the first transistor may more sufficiently cover the channel, so that the control ability of the gate is stronger.

In the embodiment of the present disclosure, the first transistor may be used as the access transistor of a dynamic memory. In this way, one of the source and the drain of the first transistor needs to be electrically connected with a capacitor, that is, the top of the semiconductor pillar 01 needs to be electrically connected with the capacitor. The covering layer 20 is made of an insulating material, so that unnecessary electrical connections may be avoided, thereby avoiding short circuit. Then, a plurality of through holes may be formed on the covering layer 20, the position of each through hole corresponds to one semiconductor pillar 01, and the through hole may expose the top of the semiconductor pillar 01. A part of an electrode plate of the capacitor may be formed by covering the inner wall of the through hole, so that the top of the semiconductor pillar 01 is electrically connected with the capacitor. Since the material of the covering layer 20 includes the boron-containing compound, the covering layer 20 is easier to be etched than the common insulating dielectric layers (such as silicon nitride, silicon oxide, etc.), and the internal structure of the through hole is more likely to show an isotropic etching result. Here, isotropy means that the etching direction may be performed in all directions, not only in a single direction. That is, for the through hole showing the isotropic etching result, the cross-sectional profile of the inner wall of the through hole may not only include straight lines, but irregular lines such as arcs, so that the area of the inner wall of the through hole may be larger. In this way, on the one hand, the contact area between the capacitor and the semiconductor pillar 01 may be increased, and the contact resistance may be reduced, and on the other hand, the surface area of the electrode plate of the capacitor be increased, and the capacitance may be increased. FIGS. 2 and 3 illustrate an optional embodiment of a through hole formed on a covering layer 20 in FIG. 1. Herein, FIG. 3 is a top view, and FIG. 2 is a cross-sectional view cut along a cross-sectional line A-A1 in FIG. 3. It is to be noted that the first direction X and the second direction Y shown in FIG. 3 are both perpendicular to the vertical direction Z shown in FIGS. 1 and 2. The first direction X and the second direction Y may be perpendicular to each other, and may also be at any included angle, and the following is exemplarily illustrated in a case where the first direction X is perpendicular to the second direction Y.

In some embodiments of the present disclosure, a first interconnecting hole 201 extending in the vertical direction Z is formed on the covering layer 31. A lower electrode plate of the capacitor may be formed in the first interconnecting hole 201, and is electrically connected with the semiconductor pillar 01.

In the embodiment of the present disclosure, referring to FIG. 2, the first interconnecting hole 201 may expose the top of the semiconductor pillar 01 and part of the sidewall close to the top. In this way, the contact area between the semiconductor pillar 01 and the subsequently formed contact electrodes is increased, thereby reducing the contact resistance and improving the electrical performance.

In the embodiment of the present disclosure, referring to FIG. 2, the middle aperture of the interconnecting hole 201 is larger than the top aperture or bottom aperture thereof. It is to be noted that the middle of the first interconnecting hole 201 refers to an area between the top and bottom thereof. Since the lower electrode plate of the capacitor may be formed in the first interconnecting hole 201, the top aperture or bottom thereof is too large, the capacitor is not easily protected, and the risk of short circuit is large. Therefore, the

top aperture or bottom aperture of the first interconnecting hole **201** may not be too large. Then, the middle aperture of the first interconnecting hole **201** is increased, so that the middle aperture is larger than the top aperture or bottom aperture thereof. In this way, on the one hand, the risk of short circuit is not increased, and on the other hand, the surface area of the electrode plate of the capacitor is increased, and the capacitance is increased.

In some embodiments of the present disclosure, as shown in FIG. 4, the top of each semiconductor pillar **01** is provided with a contact part **02**, the contact part **02** covers the top and part of the sidewall of the semiconductor pillar **01**, and the contact part **02** is configured to be in contact with the capacitor. The semiconductor pillar **01** is in electrical contact with the capacitor through the contact part **02**, which is equivalent to increasing the electrical contact area between the semiconductor pillar **01** and the capacitor, thereby reducing the contact resistance between the semiconductor pillar **01** and the capacitor. The covering layer **20** isolates the adjacent semiconductor pillars **01**, the covering layer **20** isolates the adjacent contact parts **02**, and the covering layer **20** is made of an insulating material, so that an insulating arrangement is achieved between the discrete semiconductor pillars **01**, and the contact part **02** disposed on each semiconductor pillar **01** also achieves the insulating arrangement. The covering layer **20** may form the larger through hole, so that the contact area between the contact part **02** and the subsequently formed capacitor may be increased, thereby further reducing the contact resistance and improving the performance of a semiconductor device.

The height of the sidewall of the semiconductor pillar **01** covered by the contact part **02** is $\frac{1}{12}$ - $\frac{1}{8}$ of the height of the semiconductor pillar **01**. Such a height setting may ensure that the contact part **02** and the top surface of the gate structure **10** have a sufficient safety distance, the generation of parasitic resistance between the gate structure **10** and the side of the contact part is reduced, and the conductivity of the semiconductor pillar **01** is increased at the same time.

In some embodiments of the present disclosure, referring to FIG. 1, the material of the covering layer includes a silicon boron material.

In some embodiments of the present disclosure, the silicon boron material is silicon boron nitride SiB_xN_y . Herein, $y-x \leq 2$. Here, $4 \geq y > x > 0$. It is to be noted that the representation method of SiB_xN_y does not mean that the number of the silicon atoms is 1.

In some embodiments of the present disclosure, the boron silicon material is BPSG or boro silicate glass.

Compared with general materials, the etching rate of the silicon boron materials such as silicon boron nitride, BPSG or boro silicate glass is higher, that is, the silicon boron material is easier to be etched. Therefore, the covering layer **20** made of the silicon boron material is adopted, which is easier to form the through hole with a larger area of the inner wall. Thus, on the one hand, the contact area between the capacitor and the semiconductor pillar **01** may be increased, and the contact resistance may be reduced, and on the other hand, the surface area of the electrode plate of the capacitor be increased, and the capacitance may be increased.

In some embodiments of the present disclosure, referring to FIG. 1, the top of the gate-all-around structure **101** is lower than the top of the semiconductor pillar **01**. A second part **402** of the dielectric layer **40** covers the top of the gate-all-around structure **101** and is disposed between the gate-all-around structure **101** and the covering layer **20**. The

thickness of the second part **402** of the dielectric layer **40** is greater than that of the first part **401** of the dielectric layer **40**.

FIGS. 5 and 6 illustrate some structures disposed below the covering layer in the semiconductor structure, FIG. 6 is a top view, and FIG. 5 is a cross-sectional view cut by a cross-sectional line A-A1 in FIG. 6.

In the embodiment of the present disclosure, referring to FIG. 6, in a top view, the size of the semiconductor pillars **01** meets $4F^2$ (F : the minimum pattern size obtainable under a given process condition), that is, the distance between the center points of the two adjacent semiconductor pillars **01** in the first direction X is $2F$, and at the same time, the distance between the center points of the two adjacent semiconductor pillars **01** in the second direction Y is also $2F$. Therefore, the integration density of the semiconductor structure **80** is improved. In some embodiments, when storage units are formed on the semiconductor pillar **01**, the storage units may also be disposed according to $4F^2$.

In some embodiments of the present disclosure, referring to FIGS. 5 and 6, the gate structure **10** further includes at least one bridge gate structure **102**. The at least one bridge gate structure **102** penetrates through the semiconductor pillar **01** and extends to the inner wall of the gate-all-around structure **101** in the penetrating direction. The at least one bridge gate structure **102** is disposed in the semiconductor pillar **01** and is disposed corresponding to the middle area of the gate-all-around structure **101**. In some embodiments of the present disclosure, referring to FIGS. 5 and 6, the width of the bridge gate structure **102** is less than that of the semiconductor pillar **01**. The width of the bridge gate structure **102** and the width of the semiconductor pillar **01** are the distances in the direction X .

In the embodiment of the present disclosure, in combination with FIGS. 5 and 6, the bridge gate structure **102** of the gate-all-around structure **101** penetrates through the semiconductor pillar **01** and extends to the inner wall of the gate-all-around structure **101** in the penetrating direction (i.e. the second direction Y), the semiconductor pillar **01** is penetrated to form a through hole, which is filled with the bridge gate structure **102**, that is, the inner wall of the through hole is covered with the bridge gate structure **102**. In this way, the bridge gate structure **102** covers part of the channel, the coverage area of the channel by the gate structure **10** is increased, and the channel is more easily controlled, thereby further improving the control ability of the gate of the formed first transistor.

In some embodiments of the present disclosure, as shown in FIGS. 5 and 6, the semiconductor structure **80** further includes: a first isolation layer **41**. The first isolation layer **41** is disposed between the adjacent semiconductor pillars **01**, and the gate structure **10** is disposed between the first isolation layer **41** and the semiconductor pillar **01**.

In the embodiment of the present disclosure, the first isolation layer **41** extends in the second direction Y to isolate the adjacent semiconductor pillars **01**. In addition, the gate structures **10** disposed in the second direction Y are connected to each other to form a word line extending in the direction Y , and the first isolation layer **41** is also configured to isolate the adjacent word lines **50**. The material of the first isolation layer **41** may be silicon nitride (SiN).

It can be understood that, since the gate structure **10** is a part of the word line **50**, that is, the word line **50** is in contact with the semiconductor pillar **01** through the gate structure **10**. Due to the arrangement of the gate-all-around structure **101** and the bridge gate structure **102**, the contact area between the gate structure **10** and the semiconductor

pillar **01** is increased. Therefore, the contact area between the word line **50** and the semiconductor pillar **01** is increased, and the contact resistance between the word line **50** and the semiconductor pillar **01** is reduced, thereby improving the electrical stability of the word line **50** and improving the control ability of the word line **50** to the first transistor.

FIG. 7 illustrates a cross-sectional view of the semiconductor structure **80** in a case where the number of bridge gate structures **102** is two.

In some embodiments of the present disclosure, as shown in FIG. 7, the gate structure **10** further includes at least two bridge gate structures **102**. The at least two bridge gate structures **102** are stacked at an interval in the vertical direction Z, and the at least two bridge gate structures **102** are isolated directly by the semiconductor pillar.

It can be understood that, each bridge gate structure **102** covers the inner wall of one through hole on the semiconductor pillar **01**, that is, covers a part of the channel, and the coverage area of the channel may be increased using the at least two bridge gate structures **102**, thereby increasing the control ability of the gate of the formed first transistor.

In the embodiment of the present disclosure, referring to FIG. 7, the width of each bridge gate structure **102** is smaller than that of the semiconductor pillar **01**. The semiconductor pillar **01** is penetrated by the bridge gate structure **102**, and the bridge gate structure **102** covers the inner wall of the through hole, so that the contact area between the gate structure **10** and the semiconductor pillar **01** is increased, the control ability of the gate structure **10** to the semiconductor pillar **01** is also increased, and the contact resistance between the gate structure **10** and the semiconductor pillar **01** is reduced.

In some embodiments of the present disclosure, as shown in FIG. 8, the semiconductor structure **80** further includes: a bit line **51**. The bit line **51** is disposed in the base **00**, and the bottom of the semiconductor pillar **01** is electrically connected with the bit line **51**.

In the embodiment of the present disclosure, FIG. 8 is a cross-sectional view, and FIG. 9 illustrates a positional relationship between the word line **50** and the bit line **51** in a top view. In combination with FIGS. 8 and 9, the bit line **51** extends in the first direction X, and the word line **50** extends in the second direction Y. The word line **50** is disposed above the bit line **51**. In FIG. 9, the edge of the word line **50** is represented by a solid line, the edge of the bit line **51** is represented by a dotted line, and the bit line **51** is partially shielded by the word line **50**. In a top view, the word line **50** and the bit line **51** intersect (but are not directly electrically connected), and the intersection is disposed at the position of the semiconductor pillar **01**.

Referring to FIG. 8, the base **00** may further include a metal silicide structure **03** (an area surrounded by a circular dotted line frame). The metal silicide structures **03** are connected to each other in the first direction X to form the bit line **51**. The bit line **51** is electrically connected with the bottom of the semiconductor pillar **01**.

In the embodiment of the present disclosure, the material of the metal silicide structure **03** includes at least one of cobalt silicide, nickel silicide, molybdenum silicide, titanium silicide, tungsten silicide, tantalum silicide, or platinum silicide.

It can be understood that, compared with an un-metallized semiconductor material, the metal silicide structure **03** has a relatively small resistivity, which is beneficial to reduce the resistance of the bit line **51** and reduce the contact resistance

between the bit line **51** and the semiconductor pillar **01**, thereby further improving the electrical performance of the semiconductor structure **80**.

In some embodiments of the present disclosure, as shown in FIG. 10, the semiconductor structure **80** further includes: a sacrificial structure **30** disposed above the covering layer **20**. The sacrificial structure **30** includes an additional doping area **302** and a body area **303** which are alternately disposed in the vertical direction Z. The doping concentration in the additional doping area **302** is higher than the doping concentration in the body area **303**. For example, if the material of the sacrificial structure **30** is BPSG doped with the boron element, the boron content in the additional doping area **302** is higher than the boron content in the body area **303**. Of course, in some embodiments, the body area **303** may also be un-doped BPSG.

In the embodiment of the present disclosure, an Ion Implantation (IMP) process may be used to dope the sacrificial structure **30** to form the additional doping area **302**. IMP is controlled by adopting different energies, so that areas with different depths in the sacrificial structure **30** may be doped, thereby forming the alternately disposed additional doping area **302** and body area **303**.

In some embodiments of the present disclosure, as shown in FIG. 10, the sacrificial structure **30** may be processed to form a second through hole **301**. In the subsequent process, the capacitor may be formed in the second through hole **301**, and the capacitor is electrically connected to the semiconductor pillar **01**.

In the embodiment of the present disclosure, the second through hole **301** is formed by etching according to a mask. Since the doping concentration of the boron in the additional doping area **302** is higher than the doping concentration in the body area **303**, the additional doping area **302** is easier to be etched than the body area **303**, and the additional doping area **302** is easier to form an isotropic etched structure. Referring to FIG. 7, the aperture of the second through hole **301** in the additional doping area **302** is larger than that in the body area **303**, so that the cross-sectional profile of the inner sidewall of the second through hole **301** is wavy.

It can be understood that, since the electrode plate of the capacitor formed in the second through hole **301** covers the inner sidewall of the second through hole **301**, the inner sidewall of the second through hole **301** with the wavy cross-sectional profile may increase the area of the electrode plate of the capacitor, thereby increasing the capacity of the capacitor. At the same time, the first interconnecting hole **201** may expose the top of the semiconductor pillar **01** and part of the sidewall close to the top, so that the contact area between the semiconductor pillar **01** and the capacitor is increased, thereby reducing the contact resistance and improving the electrical performance.

The embodiments of the present disclosure further provide a method for manufacturing a semiconductor structure, which includes S101 to S104 and may be described in combination with each step.

It is to be noted that, FIGS. 11 to 20 are partial structural diagrams of a semiconductor structure in each step, which are configured to describe and clearly illustrate the steps of the method for manufacturing the semiconductor structure. A first direction X and a second direction Y shown in FIGS. 11 to 20 are both perpendicular to the vertical direction Z. The first direction X and the second direction Y may be perpendicular to each other, and may also be at any included

11

angle, and the following is exemplarily illustrated in a case where the first direction X is perpendicular to the second direction Y.

At S101, a base is provided, the base is patterned, and discrete semiconductor pillars are formed on the base.

In the embodiment of the present disclosure, as shown in FIG. 11, after the base 00 is patterned, the discrete semiconductor pillars 01 are formed. The semiconductor pillars 01 are disposed at the top of the base 00 and extend in a vertical direction Z. The base 00 may include at least one of semiconductor materials, for example, group IV elements such as silicon (Si), germanium (Ge), and silicon germanium (SiGe), or group III-V components such as gallium arsenide (GaAs), aluminum gallium arsenide (AlGaAs), indium arsenide (InAs), or indium gallium arsenide (InGaAs), and the following is exemplarily illustrated with the silicon element included in the base 00.

The semiconductor pillar 01 may have a doping element to improve the conductivity of the semiconductor pillar 01. Herein, the doping element may be a P-type doping element or an N-type doping element, and the N-type doping element may be at least one of an arsenic (As) element, a phosphorus (P) element, or an antimony (Sb) element. The P-type doping element may be at least one of a boron (B) element, an indium (In) element, or a gallium (Ga) element.

In the embodiment of the present disclosure, referring to FIG. 11, the base 00 is further provided with a bit line 51, and the bit line 51 extends in the first direction X. The base 00 further includes a metal silicide structure 03 (an area surrounded by a circular dotted line frame). The metal silicide structures 03 are connected to form the bit line 51. The bottom of the semiconductor pillar 01 is electrically connected with the bit line 51. The material of the metal silicide structure 03 includes at least one of cobalt silicide, nickel silicide, molybdenum silicide, titanium silicide, tungsten silicide, tantalum silicide, or platinum silicide.

Referring to FIG. 11, the semiconductor pillar 01 is also covered with a mask layer 60, a first part 601 of the mask layer 60 covers the sidewall and the top of the semiconductor pillar 01, and the middle of the first part 601 is filled with a second part 602 of the mask layer 60. The materials of the first part 601 and the second part 602 of the mask layer 60 are different, for example, the material of the first part 601 is silicon oxide, and the material of the second part 602 is silicon nitride. In this way, in combination with FIGS. 12 and 13, etching may be performed according to a certain selection ratio, only the second part 602 and part of the semiconductor channel 01 covered therewith are removed, and a groove 61 is formed in the middle of the semiconductor channel 01.

FIG. 14 is a top view of the single semiconductor pillar 01. As shown in FIG. 14, the groove 61 extends in the second direction Y and penetrates through the semiconductor pillar 01.

In the embodiment of the present disclosure, referring to FIG. 12, before the groove is formed by etching, a first isolation layer 41 may be deposited. The first isolation layer 41 extends in the second direction Y to isolate the adjacent semiconductor pillars 01. The material of the first isolation layer 41 may be silicon nitride.

In the embodiment of the present disclosure, in combination with FIGS. 13 and 15, after the groove 61 is formed by etching, an epitaxial layer 62 may be formed in the inner wall of the groove 61 to reduce the width of the groove 61, and repair the defects of the inner wall of the groove 61.

In the embodiment of the present disclosure, in combination with FIGS. 15 and 16, after the groove 61 is formed,

12

a sacrificial layer 63 may be formed in the groove 61. The height of the sacrificial layer 63 is $\frac{1}{3}$ - $\frac{1}{2}$ of the height of the groove 61. The material of the sacrificial layer 63 may be silicon germanium (SiGe), which is easily removed in the subsequent process, and provides a basis for forming a bridge gate structure in the gate structure in the subsequent process. Then, referring to FIG. 16, a filling layer 64 may be formed in the groove 61, the filling layer 64 is disposed on the sacrificial layer 63, and the remaining groove 61 is filled with the filling layer 64.

In the embodiment of the present disclosure, in combination with FIGS. 12 and 17, the part, covering the sidewall of the semiconductor pillar 01, of the first part 601 of the mask layer 60 may be etched to the bottom area of the semiconductor pillar 01. As shown in 17, in this way, the remaining first part 601 may protect the sidewall, close to the bottom, of the semiconductor pillar 01 may be protected to avoid subsequent short circuit between the word line and the bit line.

At S102, a dielectric layer is formed.

In the embodiment of the present disclosure, referring to FIGS. 17 and 18, a wet etch process is adopted to remove the sacrificial layer 63 filled in the semiconductor pillar 01, so as to form the through hole 013 in FIG. 18.

In the embodiment of the present disclosure, referring to FIGS. 18 and 19, after the through hole 013 is formed, a Chemical Vapor Deposition (CVD) process may be adopted to form a first initial dielectric layer on the sidewall of the middle area of the semiconductor pillar 01 and the inner wall of the through hole 013. Herein, the part of the first initial dielectric layer covering the semiconductor pillar 01 with reduced size forms a first part 401 of the dielectric layer 40, and there is a gap between the first part 401 of the dielectric layer 40 and the first isolation layer 41 to form the gate layer of a gate-all-around structure 101.

Then, in combination with FIGS. 18 and 19, after the gate-all-around structure 101 is formed, a second initial dielectric layer may be deposited. The part of the second initial dielectric layer covering the top of the gate-all-around structure 101 forms a second part 402 of the dielectric layer 40. In this way, the dielectric layer 40 is formed. The dielectric layer 40 covers the sidewall of the semiconductor pillar 01. The thickness of the second part 402 of the dielectric layer 40 is greater than that of the first part 401 of the dielectric layer 40.

At S103, a gate structure is formed in the middle area of the semiconductor pillar.

In the embodiment of the present disclosure, in combination with FIGS. 18 and 19, after the first initial dielectric layer is formed, the gate layer is formed around the sidewall of the first initial dielectric layer through filling the through hole 013. Herein, the part of the gate layer for filling the through hole 013 forms a bridge gate structure 102. The material of the gate structure may be conductive materials such as titanium nitride, polycrystalline silicon, or tungsten. The sidewall of the first initial dielectric layer may be covered with the gate layer, and then the part of the gate layer around the sidewall of the first initial dielectric layer may be etched back. The gate layer remaining after the etchback forms the gate-all-around structure 101. In this way, the gate structure 10 including the gate-all-around structure 101 and the bridge gate structure 102 is formed. Herein, the gate-all-around structure 101 surrounds the semiconductor pillar 01, and the bridge gate structure 102 penetrates through the semiconductor pillar 01 and extends to the inner wall of the gate-all-around structure 101 in the

13

penetrating direction. The first part **401** of the dielectric layer **40** is disposed between the gate structure **10** and the semiconductor pillar **01**.

It can be understood that, the semiconductor pillar **01** is penetrated to form the through hole **013**, which is filled with the bridge gate structure **102**, that is, the inner wall of the through hole **013** is covered with the bridge gate structure **102**. In this way, the coverage area of the semiconductor pillar **01** by the gate structure **10** is increased, thereby further improving the control ability of the gate of the formed transistor.

At **S104**, a covering layer is formed.

In the embodiment of the present disclosure, referring to FIGS. **19** and **20**, first, the second part **402** of the dielectric layer **40** may be etched back, so that the top of the second part **402** of the dielectric layer **40** is lower than the top of the semiconductor pillar **01**. Then, referring to FIG. **20**, the covering layer **20** may be deposited. The covering layer **20** covers the top of the semiconductor pillar **01** and part of the sidewall close to the top. The material of the covering layer **20** includes a boron-containing compound, that is, the material of the covering layer **20** has a higher etching rate than general materials.

In the embodiment of the present disclosure, the material of the covering layer **20** may be silicon boron nitride SiB_xN_y . Herein, $4 \geq y > x > 0$, $y - x \leq 2$. In some embodiments of the present disclosure, the boron silicon nitride SiB_xN_y may be SiB_2N_4 or SiB_2N_4 . It is to be noted that the representation method of SiB_xN_y does not mean that the number of the silicon atoms is 1. Compared with the general materials, the etching rate of the silicon boron nitride is higher, that is, the silicon boron nitride is easier to be etched.

In the embodiment of the present disclosure, when the silicon nitride is formed, the boron element is introduced into a cavity, thereby forming the silicon boron nitride, that is, the covering layer **20**. At the same time, by controlling a flow ratio of the nitrogen atoms to the boron atoms, the ratio of the number of the boron atoms to the number of the nitrogen atoms in the silicon boron nitride may be adjusted. It is to be noted that, since the silicon boron nitride is easier to be etched than silicon nitride, and the boron atoms are doped into the silicon nitride at the same time, so that the stress of the silicon nitride may be reduced, that is, the stress of the silicon boron nitride is smaller, the internal interaction force is smaller, and the structural damage is not prone to generation, thereby improving the device performance. When the difference between the number of the nitrogen atoms and the number of the boron atoms is smaller than or equal to 2, the content of the nitrogen atoms in the silicon boron nitride may be increased, thereby increasing the etching rate of the silicon boron nitride, and the stress of the silicon boron nitride is smaller.

It can be understood that, on the one hand, since the material of the covering layer **20** has a higher etching rate than the general materials, the covering layer **20** is etched to be easier to form the through hole with a larger middle aperture, and the top of the semiconductor pillar **01** and part of the sidewall close to the top are exposed, which may refer to FIGS. **2** and **3**. Therefore, for the capacitor formed in the through hole, the surface area of the electrode plate is larger, and the capacitance is larger, and at the same time, the contact area with the semiconductor pillar **01** is larger, and the contact resistance is smaller.

In some embodiments of the present disclosure, in combination with FIGS. **12** to **19**, the step of forming the gate structure **10** includes **S201** to **S207**, which may be described in combination with each step.

14

At **S201**, a groove **61** is formed in the semiconductor pillar **01**.

In the embodiment of the present disclosure, as shown in FIG. **12**, the semiconductor pillar **01** is also covered with a mask layer **60**, a first part **601** of the mask layer **60** covers the sidewall and the top of the semiconductor pillar **01**, and the middle of the first part **601** is filled with a second part **602** of the mask layer **60**.

In combination with FIGS. **12** and **13**, etching may be performed according to a certain selection ratio, only the second part **602** and part of the semiconductor pillar **01** covered therewith are removed, and a groove **61** is formed in the middle of the semiconductor pillar **01**. FIG. **14** is a top view of the single semiconductor pillar **01**. As shown in FIG. **14**, the groove **61** extends in the second direction **Y** and penetrates through the semiconductor pillar **01**.

In the embodiment of the present disclosure, in combination with FIGS. **13** and **15**, after the groove **61** is formed by etching, an epitaxial layer **62** may be formed in the inner wall of the groove **61** to reduce the width of the groove **61**. Since the smaller the size of a mask pattern, the greater the difficulty of the process, the etching is first performed according to the second part **602** of the wider mask layer, and then the epitaxial layer **62** is formed to reduce the width of the groove **61**. In this way, a small-size groove is formed with a large-size mask pattern, which reduces the difficulty of the process. At the same time, in the process of forming the groove **61** by etching, defects and damages may be brought to the inner wall of the groove **61**, and the epitaxial layer **62** may also repair the defects and damages on the inner wall of the groove **61**, thereby reducing the defects of the semiconductor pillar **01** and improving the performance of the formed semiconductor structure.

At **S202**, a sacrificial layer **63** is formed in the groove **61**.

In the embodiment of the present disclosure, in combination with FIGS. **15** and **16**, after the groove **61** is formed, a sacrificial layer **63** may be formed in the groove **61**. The material of the sacrificial layer **63** may be silicon germanium (SiGe), which is easily removed in the subsequent process.

At **S203**, a filling layer **64** is formed in the groove **61**, the filling layer **64** is disposed on the sacrificial layer **63**, and the remaining groove **61** is filled with the filling layer **64**.

In the embodiment of the present disclosure, if a gate structure to be formed only includes one bridge gate structure, after the sacrificial layer **63** is formed, the filling layer **64** is formed in the groove **61**, the filling layer **64** is disposed on the sacrificial layer **63**, and the remaining grooves **61** is directly filled with the filling layer **64**, as shown in FIG. **16**.

If the gate structure to be formed includes at least two bridge gate structures, after the sacrificial layer **63** is formed, part of the sacrificial layer **63** is etched back to form the filling layer **64** in the groove **61**, and the filling layer **64** is disposed on the remaining sacrificial layer **63**. That is, after one filling layer **64** is formed on the sacrificial layer **63**, the sacrificial layer **63** and the filling layer **64** are formed in the groove **61** in sequence, and finally the top of the groove **61** is filled with the filling layer **64**.

It can be understood that the sacrificial layer **63** and the filling layer **64** are formed in the groove **61** in sequence, which provides a basis for forming the bridge gate structure in the gate structure in the subsequent process.

At **S204**, the sacrificial layer **63** is removed to form a through hole **013**.

In the embodiment of the present disclosure, referring to FIGS. **17** and **18**, a wet etch process is adopted to remove the sacrificial layer **63** filled in the semiconductor pillar **01**, so

15

as to form the through hole **013** in FIG. **18**. The through hole **013** penetrates through the semiconductor pillar **01** in the second direction Y.

At **S205**, a first initial dielectric layer is formed on the sidewall of the semiconductor pillar **01** and the inner wall of the through hole **013**.

In the embodiment of the present disclosure, in combination with FIGS. **18** and **19**, after the through hole **013** is formed, a CVD process may be adopted to form the first initial dielectric layer on the sidewall of the middle area of the semiconductor pillar **01** and the inner wall of the through hole **013**. Herein, the part, disposed in the middle area, of the first initial dielectric layer forms the first part **401** of the dielectric layer **40**, and the first part **401** of the dielectric layer **40** is disposed between the gate structure **10** and the semiconductor pillar **01**. The material of the first initial dielectric layer may be silicon oxide.

At **S206**, a gate layer is formed around the sidewall of the first initial dielectric layer through filling the through hole **013**. Herein, the part of the gate layer for filling the through hole **013** forms a bridge gate structure **102**.

In the embodiment of the present disclosure, in combination with FIGS. **18** and **19**, after the first initial dielectric layer is formed, the gate layer is formed around the sidewall of the first initial dielectric layer through filling the through hole **013**. Herein, the part of the gate layer for filling the through hole **013** forms a bridge gate structure **102**. The material of the gate structure may be conductive materials such as titanium nitride.

At **S207**, the part of the gate layer around the sidewall of the first initial dielectric layer is etched to form a gate-all-around structure **101**.

In the embodiment of the present disclosure, referring to FIG. **19**, after the gate layer is formed, the part of the gate layer around the sidewall of the first initial dielectric layer may be etched back, and the gate layer remaining after the etchback forms the gate-all-around structure **101**. In this way, the gate structure **10** including the gate-all-around structure **101** and the bridge gate structure **102** is formed.

It can be understood that, the gate structure **10** includes the gate-all-around structure **101** and the bridge gate structure **102**, and the coverage area of the semiconductor pillar **01** is increased, thereby further improving the control ability of the gate of the formed transistor.

In the embodiment of the present disclosure, in combination with FIGS. **18** to **19**, before the first initial dielectric layer is formed, the method for manufacturing the semiconductor structure further includes **S208**, which may be illustrated in combination with each step.

At **S208**, part of the sidewall of the semiconductor pillar **01** is etched, so that the width of the middle area of the semiconductor pillar **01** is smaller than the width of the bottom of the semiconductor pillar **01**.

In the embodiment of the present disclosure, before the first initial dielectric layer is formed, the area of the semiconductor pillar **01** not covered with the mask layer **601** in FIG. **18** and part of the sidewall above the bottom of the semiconductor pillar **01** may be etched (for example, dry etch), so that the width of the middle area of the semiconductor pillar **01** is smaller than the width of the bottom of the semiconductor pillar **01**, thereby providing a larger space for the subsequent formation of the gate structure. The above width may be the width in the first direction X.

In the embodiment of the present disclosure, in combination with FIGS. **19** to **20**, the step of forming the covering layer **20** includes **S301** to **S303**, which may be illustrated in combination with each step.

16

At **S301**, a second initial dielectric layer is formed on the gate structure **10** and the semiconductor pillar **01**.

In the embodiment of the present disclosure, referring to FIG. **19**, after the gate structure **10** is formed, the second initial dielectric layer may be deposited on the gate structure **10** and the semiconductor pillar **01**.

At **S302**, the second initial dielectric layer is etched until the top of the semiconductor pillar **01** and part of the sidewall close to the top are exposed.

In the embodiment of the present disclosure, in combination with FIGS. **19** and **20**, after the initial dielectric layer is deposited, the second initial dielectric layer may be etched until the top of the semiconductor pillar **01** and part of the sidewall close to the top are exposed, and the remaining second initial dielectric layer forms the second part **402** of the dielectric layer **40**.

At **S303**, the top of the semiconductor pillar **01** and part of the sidewall close to the top are covered to form the covering layer **20**.

In the embodiment of the present disclosure, referring to FIG. **20**, after the top of the semiconductor pillar **01** and part of the sidewall close to the top are exposed, the top of the semiconductor pillar **01** and part of the sidewall close to the top may be covered to deposit the covering layer **20**. The material of the covering layer **20** includes a boron-containing compound, that is, the material of the covering layer **20** has a higher etching rate than general materials.

In some embodiments of the present disclosure, referring to FIGS. **19** and **21** to **25**, before the covering layer **20** is formed, a contact part **02** may also be formed on the top surface of the semiconductor pillar **01** and the side surface close to the top surface.

Referring to FIGS. **19** and **21**, after the dielectric layer **40** is formed, a Wet Etch process or a Dry Etch process may be adopted to remove part of the dielectric layer **40** on the top of the semiconductor pillar **01** to expose the semiconductor pillar **01**, so that the second part **402** of the remaining dielectric layer **40** is flush with the top of the semiconductor pillar **01**.

Then, referring to FIGS. **21** and **22**, the wet etch process may be adopted to remove part of the top of the semiconductor pillar **01**, so that the top end surface of the semiconductor pillar **01** is lower than the top end surfaces of the dielectric layer **40** and the first isolation layer **41**. Herein, the removal height of the semiconductor pillar **01** is $\frac{1}{12}$ – $\frac{1}{8}$ of the remaining semiconductor pillar.

Then, referring to FIGS. **22** and **23**, the top ends of the semiconductor pillar **01**, the dielectric layer **40** and the first isolation layer **41** may be covered with a conductive layer **70**.

Then, referring to FIGS. **23** and **24**, the conductive layer **70** may be subjected to a grinding process until the top ends of the dielectric layer **40** and the first isolation layer **41** are exposed, so that the remaining conductive layer on the semiconductor pillar **01** forms the contact part **02**. The contact part **02** may be made of a metal material, such as titanium nitride or tungsten, so as to reduce the contact resistance between the semiconductor pillar **01** and the capacitor.

Referring to FIGS. **24** and **25**, after the contact part **02** is formed, part of the dielectric layer **40** may be removed by a wet etch process, so that the top of the second part **402** of the dielectric layer **40** is lower than the top of the semiconductor pillar **01**. Then, the covering layer **20** may be deposited. The covering layer **20** covers the top of the semiconductor pillar **01** and part of the sidewall close to the top.

17

It can be seen that the embodiments of the present disclosure provide a semiconductor structure and a method for manufacturing the same. The semiconductor structure includes: a base, gate structures, a dielectric layer, and a covering layer. Herein, the base includes discrete semiconductor pillars. The semiconductor pillars are disposed at the top of the base and extend in a vertical direction. The dielectric layer covers the sidewalls of the semiconductor pillars. The gate structures are disposed in the middle area of the semiconductor pillars. The gate structure includes a gate-all-around structure, the gate-all-around surrounding the semiconductor pillar. A first part of the dielectric layer is disposed between the gate structures and the semiconductor pillars. The covering layer covers the top of the semiconductor pillars and part of the sidewalls close to the top. The material of the covering layer includes a boron-containing compound. In this way, a through hole may be formed on the covering layer, and a part of an electrode plate of a capacitor may be formed in the through hole, so that the semiconductor pillar and the capacitor are electrically connected. Since the material of the covering layer includes a boron-containing compound, the covering layer is easier to be etched than a common insulating dielectric layer, so that the area of the inner wall of the through hole formed by etching may be larger. In this way, on the one hand, the contact area between the capacitor and the semiconductor pillar may be increased, and the contact resistance may be reduced, and on the other hand, the surface area of the electrode plate of the capacitor be increased, and the capacitance may be increased. Thus, the embodiments of the present disclosure improve the overall electrical performance of the semiconductor structure.

It is to be noted that, in the present disclosure, terms "include" and "contain" or any other variant thereof is intended to cover nonexclusive inclusions herein, so that a process, method, object or device including a series of elements not only includes those elements but also includes other elements which are not clearly listed or also includes elements intrinsic to the process, the method, the object or the device. Under the condition of no more limitations, an element defined by the statement "including a/an . . ." does not exclude existence of the same other elements in a process, method, object or device including the element.

The sequence numbers of the embodiments of the present disclosure are adopted not to represent superiority-inferiority of the embodiments but only for description. The methods disclosed in the several method embodiments provided in the present disclosure may be combined arbitrarily without conflict to obtain new method embodiments. The features disclosed in the several product embodiments provided in the present disclosure may be combined arbitrarily without conflict to obtain new product embodiments or device embodiments.

The above is only the specific implementation mode of the present disclosure and not intended to limit the scope of protection of the present disclosure. Any variations or replacements apparent to those skilled in the art within the technical scope disclosed by the present disclosure shall fall within the scope of protection of the present disclosure. Therefore, the scope of protection of the present disclosure shall be subject to the scope of protection of the claims.

In the embodiments of the present disclosure, since the material of a covering layer includes a boron-containing compound, the covering layer is easier to be etched than a

18

common insulating dielectric layer, so that the area of the inner wall of a through hole formed by etching may be larger. In this way, on the one hand, the contact area between the capacitor and the semiconductor pillar may be increased, and the contact resistance may be reduced, and on the other hand, the surface area of the electrode plate of the capacitor be increased, and the capacitance may be increased. Thus, the embodiments of the present disclosure improve the overall electrical performance of the semiconductor structure.

What is claimed is:

1. A semiconductor structure, comprising:

a base, comprising discrete semiconductor pillars, the semiconductor pillars being disposed at top of the base and extending in a vertical direction;

a dielectric layer for covering sidewalls of the semiconductor pillars;

gate structures disposed in middle area of the semiconductor pillars and comprising a gate-all-around structure, wherein each gate-all-around surrounds the semiconductor pillar, and a first part of the dielectric layer is disposed between the gate structures and the semiconductor pillars; and

a covering layer for covering top of the semiconductor pillars and part of the sidewalls close to the top, wherein material of the covering layer comprises a boron-containing compound;

wherein the gate structure further comprises at least one bridge gate structure;

the at least one bridge gate structure penetrates through the semiconductor pillar and extends to the inner wall of the gate-all-around structure in the penetrating direction; and

the at least one bridge gate structure is disposed in the semiconductor pillar and is disposed corresponding to middle area of the gate-all-around structure.

2. The semiconductor structure of claim 1, wherein the material of the covering layer comprises a silicon boron material.

3. The semiconductor structure of claim 2, wherein the silicon boron material is silicon boron nitride SiB_xN_y , where $y-x \leq 2$.

4. The semiconductor structure of claim 2, wherein the boron silicon material is Boro Phospho Silicate Glass (BPSG) or boro silicate glass.

5. The semiconductor structure of claim 1, wherein width of each bridge gate structure is less than width of the semiconductor pillar.

6. The semiconductor structure of claim 1, wherein the gate structure further comprises at least two bridge gate structures, the at least two bridge gate structures are stacked at an interval in the vertical direction, and the at least two bridge gate structures are isolated directly by the semiconductor pillar.

7. The semiconductor structure of claim 1, wherein top of the gate-all-around structure is lower than the top of the semiconductor pillar; and

a second part of the dielectric layer covers the top of the gate-all-around structure and is disposed between the gate-all-around structure and the covering layer; and thickness of the second part of the dielectric layer is greater than thickness of the first part of the dielectric layer.

8. The semiconductor structure of claim 1, further comprising: a first isolation layer;

19

the first isolation layer is disposed between adjacent semiconductor pillars, and the gate structure is disposed between the first isolation layer and the semiconductor pillar.

9. The semiconductor structure of claim 1, further comprising: a bit line;

the bit line is disposed inside the base; and bottom of the semiconductor pillar is connected with the bit line.

10. A method for manufacturing a semiconductor structure, comprising:

providing a base, patterning the base, and forming discrete semiconductor pillars on the base, the semiconductor pillars being disposed at top of the base and extending in a vertical direction;

forming a dielectric layer, the dielectric layer covering sidewalls of the semiconductor pillars;

forming gate structures in middle area of the semiconductor pillars, the gate structure comprising a gate-all-around structure, the gate-all-around surrounding the semiconductor pillar, and a first part of the dielectric layer being disposed between the gate structures and the semiconductor pillars; and

forming a covering layer, the covering layer covering top of the semiconductor pillars and part of the sidewalls close to the top, and material of the covering layer comprising a boron-containing compound,

wherein the gate structure further comprises at least one bridge gate structure;

the at least one bridge gate structure penetrates through the semiconductor pillar and extends to the inner wall of the gate-all-around structure in the penetrating direction; and

the at least one bridge gate structure is disposed in the semiconductor pillar and is disposed corresponding to middle area of the gate-all-around structure.

11. The manufacturing method of claim 10, wherein the gate structure further comprises one bridge gate structure, and the forming the gate structure comprises:

forming a groove in the semiconductor pillar; forming a sacrificial layer in the groove; and forming a filling layer is formed in the groove, the filling layer being disposed on the sacrificial layer, and remaining groove being filled with the filling layer.

12. The manufacturing method of claim 10, wherein the gate structure further comprises at least two bridge gate structures, and the forming the gate structure comprises:

20

forming a groove in the semiconductor pillar;

forming a sacrificial layer in the groove;

etching back part of the sacrificial layer, and forming a filling layer in the groove, the filling layer being disposed on remaining sacrificial layer; and

continuing to form the sacrificial layer and the filling layer in the groove in sequence until top of the groove is filled with the filling layer.

13. The manufacturing method of claim 11, wherein the forming the gate structure further comprises:

removing the sacrificial layer to form a through hole, the through hole penetrating through the semiconductor pillar;

forming a first initial dielectric layer on sidewall of the semiconductor pillar and the inner wall of the through hole, wherein part, disposed in the middle area of the semiconductor pillar, of the first initial dielectric layer forms the first part of the dielectric layer;

forming a gate layer around sidewall of the initial dielectric layer and filling the through hole, wherein part of the gate layer for filling the through hole forms a bridge gate structure; and

etching part of the gate layer around the sidewall of the first initial dielectric layer to form the gate-all-around structure.

14. The manufacturing method of claim 13, wherein after the through hole is formed and before the first initial dielectric layer is formed, the manufacturing method further comprises:

etching part of the sidewall of the semiconductor pillar, so that width of the middle area of the semiconductor pillar is smaller than width of bottom of the semiconductor pillar.

15. The manufacturing method of claim 10, wherein the forming the covering layer comprises:

forming a second initial dielectric layer on the gate structure and the semiconductor pillar,

etching the second initial dielectric layer until the top of the semiconductor pillars and part of the sidewalls close to the top are exposed, wherein remaining second initial dielectric layer forms a second part of the dielectric layer; and

covering the top of the semiconductor pillars and the part of the sidewalls close to the top to form the covering layer.

* * * * *